

Tarannum Binte Tariq

Prospective PhD Applicant in Artificial Intelligence, Machine Learning, NLP, Computer Vision, and Responsible Generative AI

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Research Profile

M.Sc. student in Information and Communication Engineering at Bangladesh University of Professionals (BUP), with a B.Sc. CGPA of 3.81/4.00 and research experience in transformer-based NLP, multilingual text classification, computer vision, medical imaging, diffusion models, and responsible generative AI. Research interests focus on trustworthy and socially impactful AI for low-resource languages, mental health analysis, online fraud detection, retained foreign object detection, clinical decision support, and cultural fairness in text-to-image generative models.

Research Interests

- Machine Learning, Natural Language Processing, Computer Vision, Multimodal AI, and Responsible/Trustworthy AI
- Transformer-based NLP, multilingual Bangla/Banglish text processing, mental health text analysis, and online fraud detection
- Diffusion models, text-to-image generative AI, cultural bias evaluation, fairness, and model interpretability
- Object detection, medical image analysis, retained foreign object detection, vision transformers, and AI for social good

Education

Master of Science (M.Sc.) in Information and Communication Engineering 2025–Present
Bangladesh University of Professionals (BUP), Dhaka, Bangladesh

- **Proposed M.Sc. Thesis / Research Focus:** AI-Based Detection and Prevention of Retained Foreign Objects in Surgical Procedures through Computer Vision, Medical Imaging, and Responsible Clinical Decision Support Systems.
- **Status:** Coursework completed; thesis research ongoing.
- **Expected Completion:** December, 2026

Bachelor of Science (B.Sc.) in Information and Communication Engineering 2021–2025
Bangladesh University of Professionals (BUP), Dhaka, Bangladesh

- **CGPA:** 3.81/4.00
- **Relevant Coursework:** Artificial Intelligence, Machine Learning, Data Mining, Cloud Computing, Internet of Things, Database Management Systems, Digital Signal Processing, Operating Systems, Algorithms, Object-Oriented Programming, Linear Algebra, Probability and Statistics.

Publications

HAN-DistilBERT for Depression Detection from Twitter Data: Integrating Hierarchical Attention with Transformer-Based Model Dec. 2024

Saif Rahman, **Tarannum Binte Tariq**, et al.

Proceedings of the 27th International Conference on Computer and Information Technology (ICCIT 2024)

DOI: 10.1109/ICCIT64611.2024.11022022

Contribution: Hybrid HAN–DistilBERT model design, data preprocessing, training, evaluation, and result analysis.

Enhanced Twitter-Based Depression Severity Classification Using Multi-Head Attention with RoBERTa and DistilBERT Fusion Dec. 2025

Tarannum Binte Tariq, Nusrot Jahan, et al.

Proceedings of the 2025 IEEE 7th International Conference on Sustainable Technologies for Industry 5.0, Green University of Bangladesh

DOI: 10.1109/STI69347.2025.11367576

Contribution: Developed a hybrid HAN–DistilBERT architecture for depression detection from Twitter data, combining word-level DistilBERT attention and sentence-level hierarchical attention to achieve 96% accuracy.

July, 2026

Accepted for Vol. 04, Issue 01, July 2026 publication, Journal of Faculty of Science and Technology, Bangladesh University of Professionals

Contribution: Developed a multilingual BERT-based recruitment scam detection framework achieving 95.6% test accuracy on a 20,000-post Bangladesh job-market dataset.

Multilingual Online Recruitment Scam Detection 2025–2026

2025-2026

- Developed a multilingual BERT-based framework to identify fraudulent online recruitment posts in the Bangladesh job market.
- Worked with mixed-source online recruitment and social-platform data, focusing on employment fraud, digital safety, and low-resource language settings.

2024-2025

- Contributed to a HAN-DistilBERT architecture for depression detection and a RoBERTa-DistilBERT fusion framework with multi-head attention for severity-level classification.
- Conducted preprocessing, model training, evaluation, result analysis, and manuscript preparation for peer-reviewed AI/NLP research.

2026–Present

- Investigating cultural and social bias in diffusion-based text-to-image generative models using culturally diverse prompts and output evaluation.
- Studying default visual representations, fairness concerns, and interpretability directions for responsible multimodal/generative AI.

2026–Present

- Built a YOLO-based object detection workflow for medical images, including annotation preparation, train-validation splitting, model training, and detection-metric analysis for foreign-object localization.

Teaching Assistant, Department of Information and Communication Engineering

Jan. 2026–Present

Jan. 2026–Present

- Assist faculty with undergraduate tutorials, laboratory sessions, assignment evaluation, instructional materials, and student mentoring.
- Support students through problem-solving sessions and academic guidance in computing and engineering-related coursework.

Programming: Python, Java, C, C++, JavaScript, PHP

Machine Learning and NLP: BERT, DistilBERT, RoBERTa, Hierarchical Attention Networks, transformer-based text classification, sentiment analysis, multilingual NLP, mental health text analysis

Computer Vision and Generative AI: YOLO, object detection, Vision Transformers, diffusion models, multimodal learning, text-to-image model evaluation, responsible generative AI

Data, Experimentation, and Tools: Pandas, NumPy, PySpark, Kaggle, Jupyter Notebook, GitHub, Visual Studio Code, LaTeX, Microsoft Office

Web, Database, and SOA: MySQL, HTML5, CSS, Bootstrap, Selenium WebDriver, Playwright, automated testing

Professional Experience

Web Development Intern Trust Bank PLC, Dhaka, Bangladesh	Nov. 2024–Feb. 2025
- Improved official website interface components using HTML, CSS, Bootstrap, and PHP. - Supported responsive design, cross-browser compatibility, and enterprise-level web standards.	

Leadership and Academic Service

Vice Chair , IEEE Computer Society BUP Student Branch	2023–2025
Organized technical workshops and seminars on emerging technologies; mentored junior members in research methods and technical skills.	
Joint Secretary , BUP InfoTech Club	2023–2025
Supported academic events, technical competitions, peer-learning sessions, and collaborative student activities.	
Class Representative , Department of ICE, BUP	2022–2025
Served as a liaison between students and faculty; coordinated academic schedules, assignments, and class communication.	

Awards and Honors

Dean’s Appreciation Award, Bangladesh University of Professionals, 2023 - recognized for academic excellence with GPA 4.00/4.00 and exemplary attendance. BUP Scholarship, Bangladesh University of Professionals, 2022–2023 - merit-based scholarship for outstanding academic performance.	
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Selected Academic Projects

Movie Recommendation System using AI	Python, Pandas, NumPy
Developed a content-based and collaborative filtering recommendation engine with preprocessing and similarity computation.	
Multilingual E-Commerce Website with Payment Gateway	PHP, MySQL
Built a full-stack e-commerce platform with multilingual support, payment-gateway integration, and database-backed functionality.	
Chat Application with Group Chat and File Transfer	Java, MySQL
Designed a socket-programming application supporting real-time messaging, group chat, and file sharing.	

Languages

Bangla: Native		English: Professional working proficiency		German: Basic
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